



Review

A state-of-the-art review of the X-FEM for computational fracture mechanics

Abdelaziz Yazid*, Nabbou Abdelkader, Hamouine Abdelmadjid

Department of Civil Engineering, University of Bechar, Bechar 08000, Algeria

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ABSTRACT

This paper presents a review of the extended finite element method X-FEM for computational fracture mechanics. The work is dedicated to discussing the basic ideas and formulation for the newly developed X-FEM method. The advantage of the method is that the element topology need not conform to the surfaces of the cracks. Moreover, X-FEM coupled with LSM makes possible the accurate solution of engineering problems in complex domains, which may be practically impossible to solve using the standard finite element method.

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* Corresponding author.

E-mail address: abdelaziz970@yahoo.fr (A. Yazid).

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1. Introduction

Modeling crack growth in a traditional finite element framework is cumbersome due to the need for the mesh to match the geometry of the discontinuity. This becomes a major difficulty when treating problems with evolving discontinuities where the mesh must be regenerated at each step. Moreover, the crack tip singularity needs to be accurately represented by the approximation [1]. Due to the fact that standard finite element methods are based on piecewise differentiable polynomial approximations, they are not well suited to problems with discontinuous and/or singular solutions. Typically, finite element methods require significant mesh refinement or meshes which conform with these features to get accurate results. In response to this deficiency of standard finite element methods, extended finite elements have been developed.

The X-FEM is a numerical method to model internal (or external) boundaries such as holes, inclusions, or cracks, without requiring the mesh to conform to these boundaries. It is based on a standard Galerkin procedure, and uses the concept of partition of unity [2,3] to accommodate the internal boundaries in the discrete model. This method was originally proposed by Belytschko and Black [4]. They presented a method for enriching finite element approximations so that crack growth problems can be solved with minimal remeshing. Dolbow et al. [5,6] and Moes et al. [7] introduced a much more elegant technique by adapting an enrichment that includes the asymptotic near tip field and a Heaviside function $H(x)$. Sukumar et al. [8] extended the concepts of this method to the three-dimensional static crack modeling. Belytschko et al. [9] unified the modeling of functions with arbitrary discontinuities and discontinuous derivatives in finite elements.

In contrast to the element enrichment schemes of Benzley [10], the accuracy of this enrichment based on a partition of unity is almost independent of element size for a large range. In addition, the use of crack tip elements requires transition elements and the technique tends to deteriorate as the element size near the crack tip decreases. About the only drawback of the present method is the need for a variable number of degrees of freedom per node.

2. General review

The extended finite element method has obtained so promising results that some authors have immediately foreseen the opportunities of applying X-FEM to many kinds of problems in which discontinuities and moving boundaries have to be modeled. The X-FEM coupled with LSM makes possible the accurate solution of engineering problems in complex domains, which may be practically impossible to solve using the standard finite element method.

2.1. Multiple cracks and crack nucleation

Daux et al. [11] applied the X-FEM to voids and to more complex geometries such as cracks with multiple branches. This facilitates the accurate modeling of interactions between cracks and holes, or systems with cracks emanating from holes. Budyn et al. [12] described a method for modeling the evolution of multiple cracks in the framework of the extended finite element method (X-FEM). Mariano and Stazi [13] analyzed the interaction between a macro-crack and a population of micro-cracks by adapting the X-FEM method to a multi-field model of micro-cracked bodies.

Bellec and Dolbow [14] exposed a note on enrichment functions for modeling crack nucleation. They focused on the particular case where the extent of the crack approaches the support size of the nodal shape functions. Remmers et al. [15] studied the possibility of defining cohesive segments that can arise at arbitrary locations and in arbitrary directions and thus allow for the resolution of complex crack patterns including crack nucleation at multiple locations, followed by growth and coalescence.

2.2. Holes and inclusion

Sukumar et al. [16] presented a methodology to modeled arbitrary holes and material interfaces in two-dimensional linear elastostatics without meshing the internal boundaries. Legrain et al. [17] studied the stability of incompressible formulations enriched with X-FEM. They focused on the application of this technique the treatment of holes, material inclusions and cracks in the incompressible limit. Sukumar and al. [18] presented a two-dimensional numerical model of micro-structural effects in brittle fracture, with an aim towards the understanding of toughening mechanisms in polycrystalline materials such as ceramics.

2.3. Graded materials

Dolbow and Nadeau [19] exposed some fundamental theoretical and numerical issues concerning the use of effective properties for the failure analysis of micro-structured materials, with a focus on functionally graded materials. Then, Dolbow

and Gosz [20] described a new interaction energy integral method for the computation of mixed-mode stress intensity factors at the tips of arbitrarily oriented cracks in functionally graded materials FGMs. Menouillard et al. [21] presented a general method for the calculation of the various stress intensity factors in functionally graded materials FGMs. Comi and Stefano Mariani [22] exposed an extended finite element simulation of quasi-brittle fracture in functionally graded materials. An ad hoc extended finite element formulation, accounting for material gradation, was developed to follow the propagation of cohesive cracks in the graded medium.

2.4. Material interfaces

Belytschko [23] proposed a simplified method for modeling solid objects by structured finite elements. The method uses implicit functions to describe the outside surface of the object and any inner surfaces, such as material interfaces, sliding surfaces and cracks. Nagashima et al. [24] applied the X-FEM method to the two-dimensional elastostatic bi-material interface cracks problem. They used asymptotic solution of a homogeneous (not interface) crack to enrich the crack tip nodes, and adopted a fourth order Gauss integration for a 4-node isoparametric element with enriched nodes. Liu et al. [25] presented a direct evaluation of mixed mode SIFs by X-FEM method in homogeneous and bi-materials. They improved the accuracy of the crack-tip displacement field and determined mixed mode SIFs directly by taking into account higher order terms of the asymptotic crack-tip displacement field for homogeneous materials as well as for bi-materials. Sukumar et al. [26] proposed a partition of unity enrichment techniques for bi-material interface cracks. The crack-tip enrichment functions are chosen as those that span the asymptotic displacement fields for an interfacial crack. The stress intensity factors for bi-material interfacial cracks were numerically evaluated using the domain form of the interaction integral. Hettich and Ramm [27] exposed a mechanical modeling of material interfaces and interfacial cracks by the X-FEM method.

Asadpoure et al. [28,29], and Asadpoure and Mohammadi [30] developed three independent sets of orthotropic enrichment functions for X-FEM analysis of crack in orthotropic media. Piva et al. [31] further extended the orthotropic crack tip solutions to elastodynamic problems. Yan and Park [32] presented a study on the application of the X-FEM to the simulation of crack growth in layered composite structures, with particular emphasis on the X-FEM's capability in predicting the crack path in near-interfacial fracture.

2.5. Crack cohesive

Wells and Sluys [33] combined the X-FEM method with the cohesive zone model to investigate fracture of concrete materials and obtained excellent mixed-mode crack prediction compared with experiments. Moes and Belytschko [34] extend the X-FEM method to the case of crack growth involving a cohesive law on the crack faces and study its performance. Zi and Belytschko [35] developed a new enrichment technique by which curved cracks can be treated with higher order enrichments. Mariani and Perego [36] proposed a numerical methodology to simulate quasi-static cohesive bi-dimensional crack propagation in quasi-brittle materials.

Xiao et al. [37] presented an incremental-secant modulus iteration scheme and stress recovery for the simulation of cracking process in quasi-brittle materials described by cohesive crack models whose softening law is composed of linear segments. Meschke and Dumstor [38] proposed a variational format of the X-FEM for propagation of cohesionless and cohesive cracks in brittle and quasi-brittle solids. Cox [39] exposed an extended finite element method with analytical enrichment for cohesive crack modeling.

2.6. Contact and friction

Dolbow et al. [40] developed a technique for the finite element modeling of crack growth with frictional contact on the crack faces. They studied fracture in 2D crack growth under three different interfacial constitutive laws on the crack faces: perfect contact and unilateral contact with or without friction. Vitali and Benson [41] developed an extended finite element method for contact in Multi-Material Arbitrary Lagrangian–Eulerian (MMALE) formulations. They demonstrated that the results of this technique are better than those obtained with the mixture theories and agree well with the Lagrangian solution. In [42], they presented a research includes the extension to friction and improvements to the accuracy and robustness of their previous study. Khoei and Nikbakht [43,44] employed an enriched finite element algorithm for numerical computation of contact friction problems. The classical finite element approximation was enriched by applying additional terms to simulate the frictional behavior of contact between two bodies. Ribeaucourt et al. [45] proposed a new fatigue frictional contact crack propagation model with the coupled X-FEM/LATIN method.

Borja [46] elucidated the concepts underlying the assumed enhanced strain (AES) and the extended FE methods with reference to frictional crack propagation simulation. Boucard et al. [47] proposed a multiscale strategy for the structural optimization of geometric details (such as holes, surface profiles, etc.) within the structures with frictional contacts. Rabczuk et al. [48] exposed a new crack tip element for the phantom-node method with arbitrary cohesive cracks. Liu and Borja [49] presented a contact algorithm for frictional crack propagation with the extended finite element method. Standard Coulomb plasticity model was introduced to model the frictional contact on the surface of discontinuity.

2.7. Dynamic modeling

Chen et al. [50] and Belytschko et al. [51] presented a new enrichment technique to avoid the difficulties encountered with the original X-FEM in time-dependent problems. Réthoré et al. [52] proposed an energy-conserving scheme for dynamic crack growth using the extended finite element method. The work proposed a generalization of the X-FEM to model dynamic fracture and time-dependent problems in a general sense, and it gave a proof of the stability of the numerical scheme in the linear case.

Menouillard et al. [53] proposed an efficient explicit time stepping for the extended finite element method (X-FEM). They carried out a numerical study of the evolution of the critical time step for elements with displacement discontinuities. Svahn et al. [54] exposed a discrete crack modeling in a new X-FEM format with emphasis on dynamic response. This work extends the theory by further incorporating arbitrary discontinuities in the approximation and presenting numerical procedures to handle several fields of discontinuities. Prabel et al. [55] demonstrated that the modelization of a propagating crack in a dynamic elastic–plastic media can be done using X-FEM linear functions approximation. Rozycki et al. [56] presented a use of the extended finite element method (X-FEM) for rapid dynamic problems. They exposed an X-FEM explicit dynamics for constant strain elements to alleviate mesh constraints on internal or external boundaries. Song and Belytschko [57] exposed a cracking node method for dynamic fracture with extended finite element (X-FEM). In the method, the growth of the actual crack is tracked and approximated with contiguous discrete crack segments that lie on finite element nodes and span only two adjacent elements.

2.8. Space–time formulation

An enriched finite element method in the context of a space–time formulation has been presented by Chessa and Belytschko [58]. The discontinuities are treated by the extended finite element method (X-FEM), which uses a local partition of unity enrichment to introduce discontinuities along a moving hyper-surface, which is described by level sets. Réthoré et al. [59] proposed a combined space–time extended finite element method. The idea of this method was to propose the use of a time partition of the unity method denoted Time extended finite element method (TX-FEM) for improved numerical simulations of time discontinuities. Chessa and Belytschko [60] presented a locally enriched space–time finite element method for solving hyperbolic problems with discontinuities. They examined how the enriched space–time formulation, in which discontinuities are explicitly tracked with enrichment and level sets, can be combined with standard finite element formulations for hyperbolic equations.

2.9. Elastic–plastic fracture

Elguedj et al. [61] proposed to use the well-known Hutchinson–Rice–Rosengren (HRR) [62,63] fields to represent the singularities in elastic–plastic fracture mechanics. This analysis is done for fatigue fracture analysis of cracks in homogeneous, isotropic, elastic–plastic two-dimensional solids subject to mixed mode in the case of confined plasticity. Several strategies of enrichment, based on the Fourier analysis results, are compared and a six-enrichment-functions basis is proposed. This new tip enrichment basis is coupled with a Newton like iteration scheme and a radial return method for plastic flow.

2.10. Intrinsic X-FEM

Fries and Belytschko [64] proposed an intrinsic extended finite element method for treating arbitrary discontinuities. Unlike the standard extended FE method, no additional unknowns are introduced at the nodes whose supports are crossed by discontinuities. The method constructs an approximation space consisting of mesh based, enriched moving least squares (MLS) functions near discontinuities and standard FE shape functions elsewhere. There is only one shape function per node, and these functions are able to represent known characteristics of the solution such as discontinuities, singularities, etc.

2.11. Superposed approaches

Lee et al. [65] exposed a combination of the extended finite element method (X-FEM) and the mesh superposition method (s-version FEM) proposed by Fish [66]. The near-tip field is modeled by superimposed quarter point elements on an overlaid mesh and the rest of the discontinuity is implicitly described by a step function on partition of unity. Xiao and Karihaloo [67] proposed an implementation of the hybrid crack element (HCE) on a general finite element mesh and in combination with the X-FEM method. They exposed the incorporation of the HCE into commercial FE packages. Nakasumi et al. [68] presented a connection of mesh superposition technique and the extended finite element method to crack propagation analysis for large scale or complicated geometry structures. Wyart et al. [69,70] provide a good description of the possible approaches that can be followed to implement X-FEM in general purpose FE software. They propose a sub-structuring approach to decompose the cracked component into safe and cracked sub-domains, which are analyzed separately by the general-purpose FE software and the extended finite element code, respectively.

2.12. Stress analysis and stress intensity calculation

Legrain et al. [71] studied the stress analysis around crack tips in finite strain problems. They showed how to solve non-linear fracture mechanics problems with the extended finite method, particularly for rubber-like materials. Béchet et al. [72] improved the robustness of the X-FEM method for stress analysis around cracks by introducing a different enrichment scheme; developing new integration quadrature for asymptotic functions and implementing a preconditioning scheme adapted to the enriched functions. Xiao and Karihaloo [73] improved the accuracy of X-FEM crack tip fields using higher order quadrature and statically admissible stress recovery. Dumstorff and Meschke [74] studied numerically the performance of the different crack propagation criteria in the framework of X-FEM based structural materials.

In several works the stress intensity factors are computed at the tip of a crack in 2D bodies using domain forms of the interaction integrals [75]. Duarte et al. [76] extracted the SIFs by a least squares fit method by minimizing the errors among the stresses calculated numerically and their asymptotic values. Nagashima et al. [24] exposed the stress intensity factor analysis of the bi-material interface crack problem. Xiao and Karihaloo [77] enhanced the accuracy of the local fields and determined the SIF directly without extra post-processing. Liu et al. [78] extend this technique to direct evaluation of mixed mode SIFs in homogeneous and bi-materials.

The X-FEM has also been utilized to model computational phenomena in areas such as fluid mechanics, phase transformations, and materials science: solidification problems [79–81], fluids mechanics [82–88], biofilms [89,90], piezoelectric materials [91] and complex industrial structures [92,93].

3. X-FEM concepts

3.1. Crack-tip enrichment

The essential idea in the extended finite element method, which is closely related to the generalized finite element method [94–97], both belonging to the class of partition of unity methods, is to add discontinuous enrichment functions to the finite element approximation using the partition of unity:

$$u(x) = \sum_{i=1}^n N_i(x) \left(u_i + \sum_{j=1}^{ne(i)} a_{ji} F_j(r, \theta) \right), \quad (1)$$

where (r, θ) is a polar coordinate system with origin at the crack tip $N_i(x)$ are the standard finite element shape functions. The enrichment coefficient a_{ji} is associated with nodes and $ne(i)$ is the number of coefficients for node i : it is chosen to be four for all nodes around the crack tip and zero at all other nodes.

The crack-tip enrichment functions in isotropic elasticity $F_i(r, \theta)$ are obtained from the asymptotic displacement fields:

$$\{F_j(r, \theta)\}_{j=1}^4 = \left\{ \sqrt{r} \sin\left(\frac{\theta}{2}\right); \sqrt{r} \cos\left(\frac{\theta}{2}\right); \sqrt{r} \sin\left(\frac{\theta}{2}\right) \sin \theta; \sqrt{r} \cos\left(\frac{\theta}{2}\right) \sin \theta \right\}. \quad (2)$$

Note that the first function in the above equation is discontinuous across the crack. It represents the discontinuity near the tip, while the other three functions are added to get accurate result with relatively coarse meshes (Fig. 1).

3.2. Heaviside function

The Heaviside jump function is a discontinuous function across the crack surface and is constant on each side of the crack: +1 on one side and –1 on the other. After intruding this jump function, the approximation will be changed to the following formula:

$$U = \sum_{i \in I} u_i N_i + \sum_{j \in J} b_j N_j H(x) + \sum_{k \in K1} N_k \left(\sum_{l=1}^4 c_k^{l1} F_l^1(x) \right) + \sum_{k \in K2} N_k \left(\sum_{l=1}^4 c_k^{l2} F_l^2(x) \right), \quad (3)$$

where

- N_i is the shape function associated to node i ,
- I is the set of all nodes of the domain,
- J is the set of nodes whose shape function support is cut by a crack,
- K is the set of nodes whose shape function support contains the crack front,
- u_i are the classical degrees of freedom (i.e. displacement) for node i ,
- b_j account for the jump in the displacement field across the crack at node j . If the crack is aligned with the mesh, b_j represent the opening of the crack,
- $H(x)$ is the Heaviside function,
- c_k^l are the additional degrees of freedom associated with the crack-tip enrichment functions F_i ,

- F_I is an enrichment which corresponds to the four asymptotic functions in the development expansion of the crack-tip displacement field in a linear elastic solid (Fig. 2).

3.3. Numerical integration

There are two difficulties for the integration of X-FEM functions: the discontinuity along the crack, and the singularity at the crack tip. The numerical integration of cut elements is generally performed by partitioning them into standard sub-elements. To get accurate results, in the sub-elements, high order Gauss quadrature rule must be used. In earlier investigations, each side of a cut element was triangulated to form a set of sub-triangles. Some authors adopt a slightly different approach, choosing instead to partition cut elements into sub-quadrilaterals (Fig. 3).

The numerical integration procedure for elements cut by the crack is as follows:

1. Build the Delaunay triangulation to get the sub-elements.
2. For each sub-elements, the coordinates and weights of Gauss points are computed and then converted into the parent coordinate system of the original element.

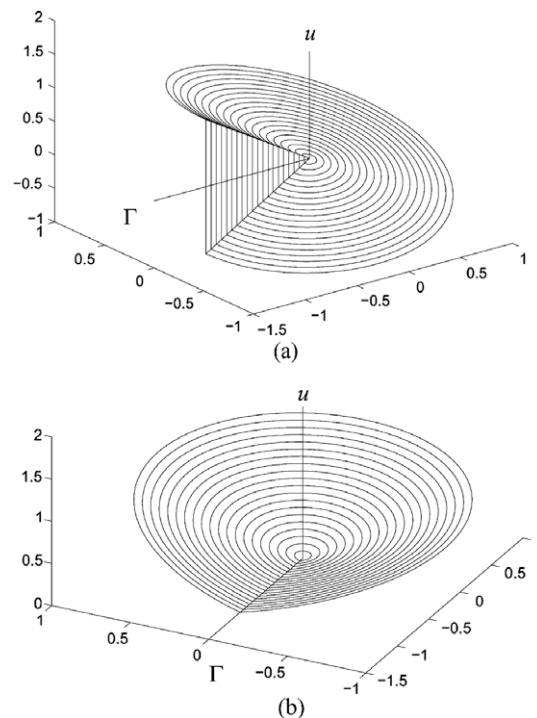


Fig. 1. Branch functions for discontinuities in: (a) the function; and (b) its derivative [9].

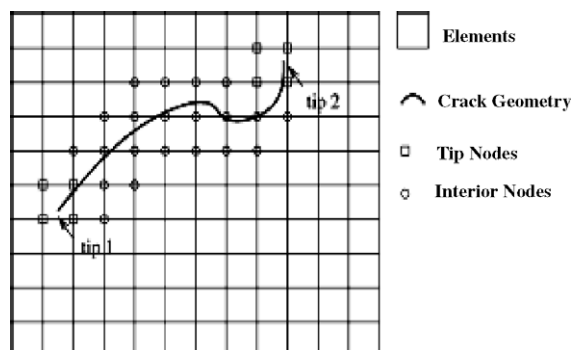


Fig. 2. A strategy of enrichment [19].

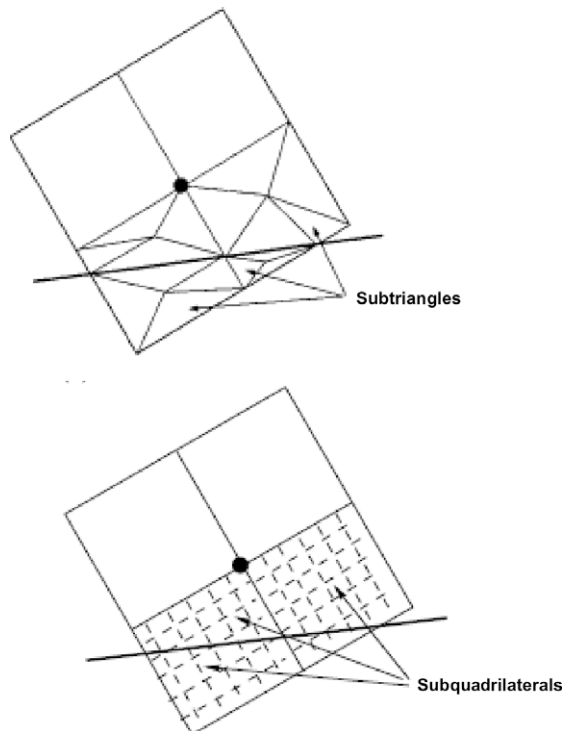


Fig. 3. Partitioning into standard sub-elements [19].

The case of removing the quadrature subcells for discontinuity functions in the X-FEM method was studied by Ventura [98]. While working, he showed how the standard Gauss quadrature could be employed in the elements with discontinuity without splitting the elements in sub-cells or to presenting any additional approximation.

3.4. Level set method

The level set method (LSM) is a numerical scheme developed by Osher and Sethian [99] to model the motion of interfaces. The principle of the method is to represent an interface by the zero of a function, called the level set function, and to update this function with Hamilton–Jacobi equations knowing the speed of the interface in the direction normal to this interface. A significant advance of the X-FEM method is given by its coupling with level set methods (LSM): the LSM is used to represent the crack location, including the location of crack tips. The X-FEM is used to compute the stress and displacement fields necessary for determining the rate of crack growth.

The X-FEM and LSM were first applied by Stolarska et al. [100] to treat crack growth in two dimensions. An algorithm coupling the LSM with X-FEM was presented to solve the elasto-static fatigue crack problem. A crack is described by two level sets:

- a normal level set, $\psi(x)$, which is the signed distance to the crack surface,
- a tangent level set $\phi(x)$, which is the signed distance to the plane including the crack front and perpendicular to the crack surface.

In a given element, $\psi_{\min}$ and $\psi_{\max}$, respectively, be the minimum and maximum nodal values of ψ on the nodes of that element. Similarly, let $\phi_{\min}$ and $\phi_{\max}$, respectively be the minimum and maximum nodal values of ϕ on the nodes of an element:

- If $\phi < 0$ and $\psi_{\min} \psi_{\max} \leq 0$, then the crack cuts through the element and the nodes of the element are to be enriched with $H(x)$.
- If in that element $\phi_{\min} \phi_{\max} \leq 0$ and $\psi_{\min} \psi_{\max} \leq 0$, then the tip lies within that element, and its nodes are to be enriched $Fi(r, \theta)$.

At point x , the radius from the crack tip and the angle of deviation from the tangent to the crack tip are given by Stolarska et al. [100]:

$$r = \sqrt{\psi^2(x, t) + \phi^2(x, t)} \quad \text{and} \quad \theta = \tan^{-1} \frac{\psi(x, t)}{\phi(x, t)}. \quad (4)$$

The X-FEM method was also employed in association with a particular level set method, the Fast Marching Method (FMM) [101] to model planar three-dimensional fatigue of crack growth. Sukumar et al. [102] have used this coupling to model simple crack growth, whereas Chopp and Sukumar [103] have studied the fatigue crack propagation of multiple coplanar cracks. In [104], Sukumar et al. presented a coupling X-FEM/FMM for non-planar three-dimensional crack growth simulations.

In Ref. [100]; two orthogonal level sets were used, one for the crack surface, the second to locate the crack tip. Ventura et al. [105] noted that one drawback of this preceding combination is that the algorithm is rendered more complicated by the need to 'freeze' the level set describing the existing crack surface. They proposed a new vector level set method for describing surfaces that are frozen behind a moving front, such as cracks. In the method, the crack geometry is described by a three-tuple for cracks in two dimensions: the sign of the level set function and the components of the closest point projection to the surface. The level set function is updated by simple geometric formulas. By contrast, in standard level set methods the level set is updated by the solution of a hyperbolic partial differential equation. Dufloy [106] presented a study of the representation of cracks with level sets. The work studies in detail the shape of the level set functions around a crack in two dimensions that is propagating with a sharp kink, obtained both with level set update methods found in the literature and with several innovative update methods developed by the author. A criterion based on the computation of a J -integral of a virtual displacement field obtained with the values of the level set functions was proposed in order to assess the quality of these update methods.

3.5. Convergence

The extended finite element method is beneficial because it provides results that are more precise than those of the classical finite element method. However, the convergence rate is not optimal with regard to the parameter of mesh. This rate is lower if compared to the one envisaged with the method of classical finite element for a slight problem as specified in Stazi et al. [107]. In the classical X-FEM method, only the nodes the nearest to the crack tip are enriched; consequently the support of the additional basis functions vanishes when the mesh parameter goes to zero. To obtain the optimal accuracy, some methods have been proposed such as X-FEM with a fixed enrichment area, high order X-FEM and the construction of blending elements [108–110]. Laborde et al. [111] have examined the capabilities of the extended finite element method efficiency for getting precise calculations in non-smooth situations such as crack problems. Some amendments were suggested to the method of X-FEM in order to get optimal accuracy. A first modification is relative to the step function of enrichment used to represent the jump in the displacement field across the crack. The second improvement is related to the asymptotic near-tip displacement solutions in the X-FEM basis.

In Ref. [111], the performances of the classical refined integration and the almost polar integration are compared by computing a X-FEM elementary matrix. Fig. 4 presents the relative error between this reference elementary matrix and a computation of the elementary matrix with the following different strategies (Figs. 5 and 6):

- using a regular refinement of the triangle, and a fixed integration rule on each refined triangle (of order 3 and 10);
- using the almost polar integration method, without any additional refinement, but for Gauss quadratures of increasing order.

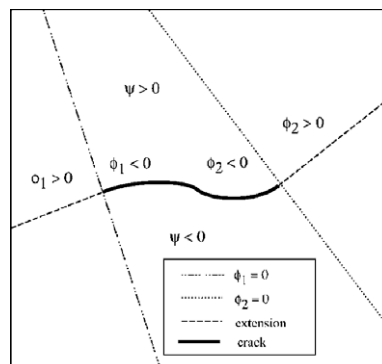


Fig. 4. Construction of initial level set functions [100].

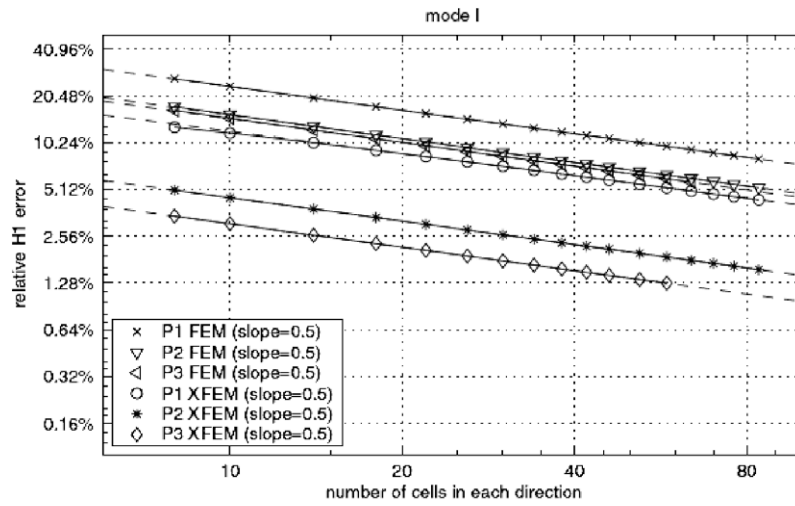


Fig. 5. Convergence comparison of classical finite elements and X-FEM for mode I problems [111].

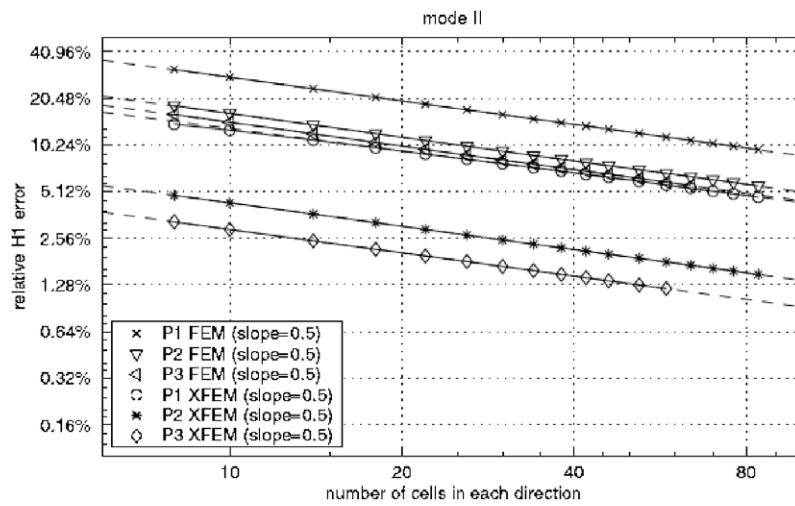


Fig. 6. Convergence comparison of classical finite elements and X-FEM for mode II problems [111].

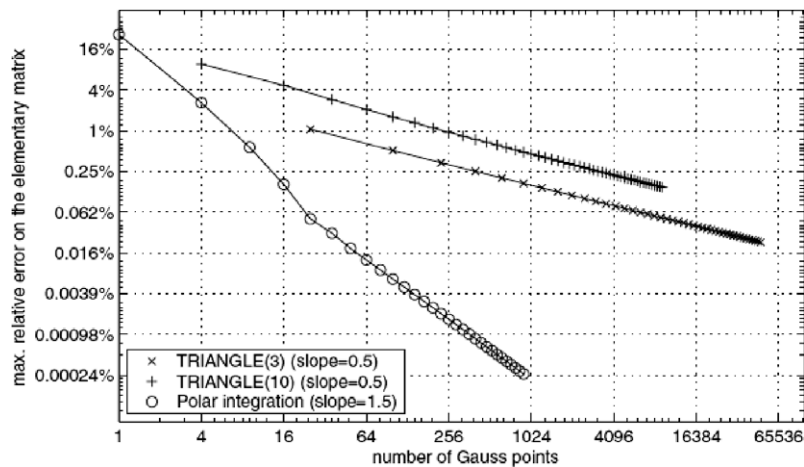


Fig. 7. Comparison of integration methods [111].

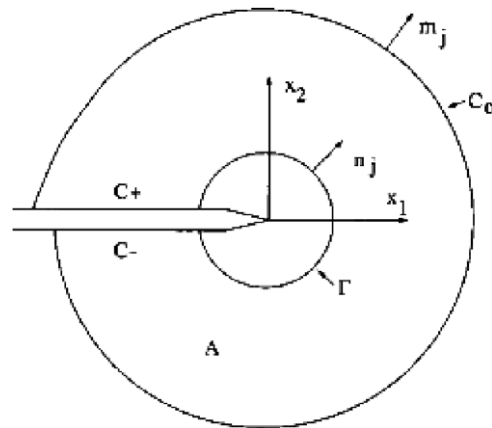


Fig. 8. Definition of the J integral around a crack.

This shows that the almost polar integration approach offers an important gain. In practice, 25 Gauss points were enough for the most accurate convergence test we have done (Fig. 7).

Chahine et al. [112,113] gave a note of the convergence result for a variant of the X-FEM method on cracked domains using a cut-off function to localize the singular enrichment area. The difficulty is caused by the discontinuity of the displacement field across the crack, but they prove that a quasi-optimal convergence rate holds in spite of the presence of elements cut by the crack. The global linear convergence rate is obtained by using an enriched linear finite element method. They proposed three improvements of X-FEM for cracked domains: The dof gathering technique, enrichment with the use of a cut-off function and non-conforming method with a bonding condition. Béchet et al. [72] exposed an improved implementation and robustness study of the X-FEM for stress analysis around cracks. They improved the robustness of the X-FEM method by three ways: (a) introducing a different enrichment scheme; (b) developing new integration quadrature for asymptotic functions and (c) implementing a preconditioning scheme adapted to the enriched functions.

3.6. Interaction integral

The stress intensity factors are computed using domain forms of the interaction integrals. According to Fig. 8, the J -integral can be defined as:

$$I^{(1,2)} = \int_{\Gamma} \left[W^{(1,2)} \delta_{1j} - \sigma_{ij}^{(1)} \frac{\partial u_i^{(2)}}{\partial x_1} - \sigma_{ij}^{(2)} \frac{\partial u_i^{(1)}}{\partial x_1} \right] n_j d\Gamma. \quad (5)$$

The contour integral (5) is not in a form best suited for finite element calculations. Moës et al. [7] recast the integral into an equivalent domain form by multiplying the integrand by a sufficiently smooth weighting function $q(x)$ which takes a value of unity on an open set containing the crack tip and vanishes on an outer prescribed contour C_0 ,

$$I^{(1,2)} = \int_A \left[\sigma_{ij}^{(1)} \frac{\partial u_i^{(2)}}{\partial x_1} + \sigma_{ij}^{(2)} \frac{\partial u_i^{(1)}}{\partial x_1} - W^{(1,2)} \delta_{1j} \right] \frac{\partial q}{\partial x_j} dA. \quad (6)$$

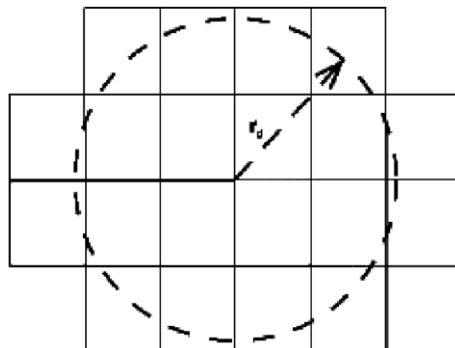


Fig. 9. Elements selected about the crack tip for calculation of the interaction integral [7].

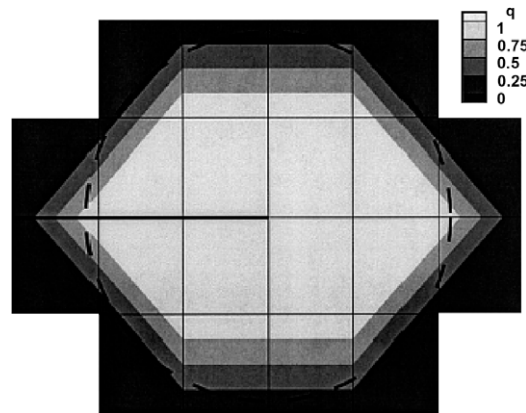


Fig. 10. Weight function q on the elements [7].

For the numerical evaluation of the above integral, the domain A is set from the collection of elements about the crack tip. They first determine the characteristic length of an element touched by the crack tip and designate this quantity as h_{local} . For two-dimensional analysis, this quantity is calculated as the square root of the element area. The domain A is then set to be all elements which have a node within a ball of radius rd about the crack tip (Figs. 9 and 10).

4. Conclusion

This article presents an overall picture and some recent progress of the extended finite element method in the crack growth modeling analysis. It sums up the significant stages carried out by the finite community element in the arena of the fracture mechanics. The study shows the use and potential of the X-FEM method to study the complex material fracture mechanisms. The method allows to make computations in a cracked domain, the crack growth being described independently of the mesh. This is an important advantage compared to existing methods where a remeshing step and an interpolation step are necessary each time an extension of the crack is computed, which can be sources of instabilities.

The extended finite element method coupled with level set method makes possible the accurate solution of engineering problems in complex domains, which may be practically impossible to solve using the standard finite element method. Moreover, by the means of the sub-structuring approach X-FEM/FEM software, the user can benefit from the many built-in features of such a code.

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